

Jae-Joon Kim

professor

基本信息

姓名: Jae-Joon Kim

所属院校: 首尔国立大学

地区: South Korea

学科领域

微电子科学与工程

集成电路设计与集成系统

人工智能

电子与计算机工程

材料物理

研究领域

深度学习与神经网络

微电子器件与制造

存储器设计与架构

低功耗电路与能量管理

薄膜材料与表面物理

低功耗设计与能耗优化

计算机体系结构

FPGA与可重构计算

数字集成电路设计

嵌入式系统与硬件设计

+3

个人简介

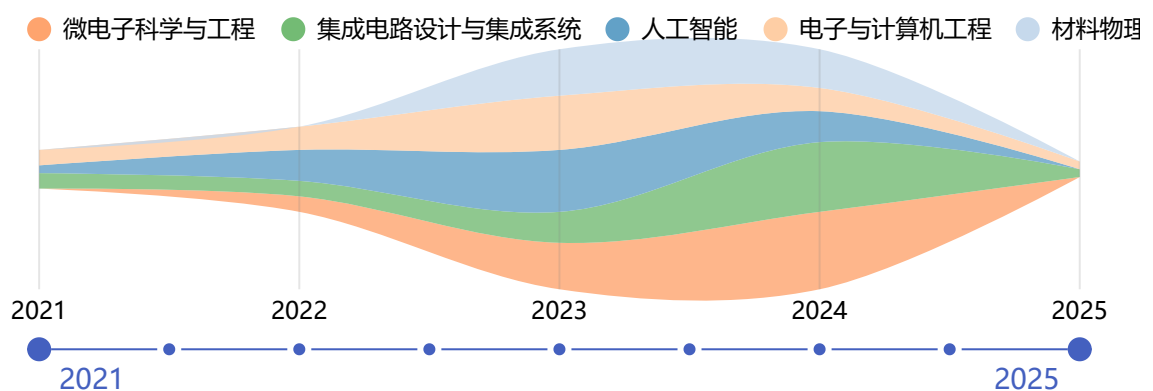
Jae-Joon Kim is a Professor at Seoul National University, specializing in VLSI Device Physics, AI HW Design, and AI Model Compression. With a robust academic profile, Professor Kim has accumulated 4,951 citations and an h-index of 36. His research collaborations span across notable institutions and industries, including Purdue University, Samsung, and IBM Research. His contributions significantly impact the field of electronics and artificial intelligence. Email: Verified email at snu.ac.kr.

工作经历

Purdue University - Professor

Sungkyunkwan University	-
SqueezeBits Inc.	-
Hanyang University - Associate Professor	-
Sogang University - Assistant Professor	-
Kinetic	-
Rebellions Inc.	-
Samsung Electronics & POSTECH	-
Kyunghee University	-
Samsung Advanced Institute of Technology	-
Pohang University of Science and Technology - Dept. of Chemical Engineering	-
Pohang Accelerator Laboratory, POSTECH	-
Amazon Web Services	-
TNO	-
Korea University - Associate Professor	-
IBM Research	-
NAVER Cloud	-
IBM T.J. Watson Research Center - Research Staff Member	-
Seoul National University - Professor	-

近年发表的论文



Real-Time Unsupervised Learning and Image Recognition via Memristive Neural Integrated Chip Based on Negative Differential Resistance of Electrochemical Metallization Cell Neuron Device

SMALL

Woo, Dae-Seong; Kim, Jae-Kyeong; Park, Gwang-Ho; Lee, Woo-Guk; Han, Min-Jong; Jin, Soo-Min; Shim, Tae-Hun; Kim, Jae-Joon; Park, Jinsub; Park, Jea-Gun

被引用次数: 0

2025

Nanoristors: highly uniform, sub-500-millivolt, large-scale, and robust molybdenum disulfide nanograined memristors

JOURNAL OF MATERIALS CHEMISTRY C

Woo, Gunhoo; Kim, Hyeong-U; Jang, Byung Chul; Naqi, Muhammad; Hong, Seongin; Bala, Arindam; Kang, Seunghun; Kim, Yunseok; Kim, Sunkook; Kim, Taesung; Kim, Jae-Joon; Yoo, Hocheon

被引用次数: 2

2024

Robust 1/f Noise Unaffected by Program/Erase Cycling-Induced Damage in Ferroelectric Schottky Barrier FETs

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 1

2024

Highly Reliable Lateral Migration-Based TFT-Type Neuron Device for Spiking Neural Networks

IEEE ELECTRON DEVICE LETTERS

Park, Min-Kyu; Hwang, Joon; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

Optimization of Programming Pulse Shape for Vertical NAND Flash Memory Using Neural Networks

IEEE ELECTRON DEVICE LETTERS

Park, Sung-Ho; Kim, Jaehyeon; Ko, Jonghyun; Im, Jiseong; Yang, Yeongheon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

On-Chip Learning in Vertical NAND Flash Memory Using Forward-Forward Algorithm

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Sung-Ho; Ko, Jonghyun; Lee, In-Seok; Koo, Ryun-Han; Kim, Jeong-Hyun; Yang, Yeongheon; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 1

2024

Reliability of FET-type gas sensor with asymmetric air-gap micro-heater structure considering thermoelectric effect

SENSORS AND ACTUATORS B-CHEMICAL

Park, Jinwoo; Jung, Gyuweon; Shin, Wonjun; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Park, Min-Kyu; Kim, Jae-Joon; Lee, Jong -Ho

被引用次数: 1

2024

Examination of Ferroelectric Domain Dynamics in HZO Under Endurance Cycling Stress

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Kim, Seungwhan; Jung, Gyuweon; Lee, Sung-Tae; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2024

Reliability Improvement in Vertical nand Flash Cells Using Adaptive Incremental Step Pulse Programming (A-ISPP) and Incremental Step Pulse Erasing (ISPE)

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Sung-Ho; Yoo, Ho-Nam; Yang, Yeongheon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 2

2024

Simultaneous Spike Processing for 3D NAND-Based Spiking Neural Networks

IEEE ELECTRON DEVICE LETTERS

Jeon, Bosung; Song, Seunghwan; Kim, Jae-Joon; Choi, Woo Young

被引用次数: 0

2024

SNNSim: Investigation and Optimization of Large-Scale Analog Spiking Neural Networks Based on Flash Memory Devices

ADVANCED INTELLIGENT SYSTEMS

Ko, Jong Hyun; Kwon, Dongseok; Hwang, Joon; Lee, Kyu-Ho; Oh, Seongbin; Kim, Jeonghyun; Im, Jiseong; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 1

2024

Multifunctional In-Memory Analog-to-Digital Converter for Next-Gen Compute-in-Memory Systems

ADVANCED INTELLIGENT SYSTEMS

Im, Jiseong; Ko, Jonghyun; Hwang, Joon; Kim, Jangsaeng; Shin, Wonjun; Koo, Ryun-Han; Park, Minkyu; Park, Sung-Ho; Choi, Woo Young; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

CMOS-Compatible Barometric Pressure Field-Effect Transistor-Type Sensor

IEEE ELECTRON DEVICE LETTERS

Lee, Chayoung; Shin, Wonjun; Jung, Gyuweon; Park, Jinwoo; Kim, Donghee; Choi, Kangwook; Shin, Hunhee; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

Mobileware: Distributed Architecture With Channel Stationary Dataflow for MobileNet Acceleration

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Ryu, Sungju; Jang, Jaeyong; Oh, Youngtaek; Kim, Jae-Joon

被引用次数: 0

2024

Lateral Migration-based Flash-like Synaptic Device for Hybrid Off-chip/On-chip Training

ADVANCED ELECTRONIC MATERIALS

Park, Min-Kyu; Hwang, Joon; Lee, Kyung Min; Woo, Sung Yun; Kim, Jae-Joon; Bae, Jong-Ho; Lee, Jong-Ho

被引用次数: 2

2024

Hardware Implementation of a Fully Functional Stochastic p-STT Neuron for Probabilistic Computing

ADVANCED ELECTRONIC MATERIALS

Jun, Han-Sol; Woo, Dae-Seong; Lee, So-Hyun; Choi, Yohan; Shin, Yeonsoo; Shim, Tae-Hun; Kim, Jae-Joon; Park, Jea-Gun

被引用次数: 1

2024

FIGNA: Integer Unit-based Accelerator Design for FP-INT GEMM Preserving Numerical Accuracy

2024 IEEE INTERNATIONAL SYMPOSIUM ON HIGH-PERFORMANCE COMPUTER ARCHITECTURE, HPCA 2024

Jang, Jaeyong; Kim, Yulhwa; Lee, Juheun; Kim, Jae-Joon

被引用次数: 1

2024

A 10T2C Capacitive SRAM-based Computing-In-Memory Macro with Array-Embedded DAC and Shift-and-Add Functions

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Kim, Eunhwan; Oh, Hyunmyung; Lee, Jehun; Park, Jihoon; Kwon, Myeongeun; Kim, Jae-Joon

被引用次数: 0

2024

Proposition of Adaptive Read Bias: A Solution to Overcome Power and Scaling Limitations in Ferroelectric-Based Neuromorphic System

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Seungwhan; Im, Jiseong; Park, Sung-Ho; Ko, Jong Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 5

2024

On-Chip Annealing Using Embedded Micro-Heater for Highly Sensitive and Selective Gas Detection

ADVANCED SCIENCE

Park, Jinwoo; Shin, Hunhee; Jung, Gyuweon; Hong, Seongbin; Park, Min-Kyu; Hwang, Joon; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 1

2024

Toward Ideal Low-Frequency Noise in Monolayer CVD MoS₂ FETs: Influence of van der Waals Junctions and Sulfur Vacancy Management

ADVANCED SCIENCE

Shin, Wonjun; Byeon, Junsung; Koo, Ryun-Han; Lim, Jungmoon; Kang, Jung Hyeon; Jang, A-Rang; Lee, Jong-Ho; Kim, Jae-Joon; Cha, Seungnam; Pak, Sangyeon; Lee, Sung-Tae

被引用次数: 0

2024

Polarization Pruning: Reliability Enhancement of Hafnia-Based Ferroelectric Devices for Memory and Neuromorphic Computing

ADVANCED SCIENCE

Koo, Ryun-Han; Shin, Wonjun; Kim, Jangsaeng; Yim, Jiyong; Ko, Jonghyun; Jung, Gyuweon; Im, Jiseong; Park, Sung-Ho; Kim, Jae-Joon; Cheema, Suraj S.; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 1

2024

Stochasticity in ferroelectric memory devices with different bottom electrode crystallinity

CHAOS SOLITONS & FRACTALS

Koo, Ryun-Han; Shin, Wonjun; Jung, Gyuweon; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 4

2024

Analysis Method of Dynamic Read Variation in a TFT-Type Synaptic Devices With Poly-Si Channel Structure

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Min-Kyu; Hwang, Joon; Yoo, Ho-Nam; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2024

A Capacitive Computing-In-Memory Circuit With Low Input Loading SRAM Bitcell and Adjustable ADC Input Range

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Eunhwan; Oh, Hyunmyung; Kang, Nameun; Park, Jihoon; Kim, Jae-Joon

被引用次数: 5

2023

Leveraging Early-Stage Robustness in Diffusion Models for Efficient and High-Quality Image Synthesis

ADVANCES IN NEURAL INFORMATION PROCESSING SYSTEMS 36 (NEURIPS 2023)

Kim, Yulhwa; Jo, Dongwon; Jeon, Hyesung; Kim, Taesu; Ahn, Daehyun; Kim, Hyungjun; Kim, Jae-Joon

被引用次数: 0

2023

Effect of Carrier Transport Process on Tunneling Electroresistance in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Dae Hwan; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 16

2023

Unveiling Resistance Switching Mechanisms in Undoped HfOx Ferroelectric Tunnel Junction Using Low-Frequency Noise Spectroscopy

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 9

2023

A Single-ring-oscillator based True-random-number- generator with 3-edges Collapse

JOURNAL OF SEMICONDUCTOR TECHNOLOGY AND SCIENCE

Kim, Eunhwan; Kim, Jae-Joon

被引用次数: 0

2023

Binaryware: A High-Performance Digital Hardware Accelerator for Binary Neural Networks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ryu, Sungju; Oh, Youngtaek; Kim, Jae-Joon

被引用次数: 0

2023

Cointegration of the TFT-Type AND Flash Synaptic Array and CMOS Circuits for a Hardware-Based Neural Network

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Min-Kyu; Kang, Won-Mook; Koo, Ryun-Han; Kim, Jeong-Hyun; Hwang, Joon; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 8

2023

Reconfigurable neuromorphic computing block through integration of flash synapse arrays and super-steep neurons

SCIENCE ADVANCES

Kwon, Dongseok; Woo, Sung Yun; Lee, Kyu-Ho; Hwang, Joon; Kim, Hyeongsu; Park, Sung-Ho; Shin, Wonjun; Bae, Jong-Ho; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 8

2023

Recovery of off-state stress-induced damage in FET-type gas sensor using self-curing method

DISCOVER NANO

Shin, Wonjun; Jeong, Yujeong; Kim, Mingyu; Lee, Jungsoo; Koo, Ryun-Han; Hong, Seongbin; Jung, Gyuweon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 2

2023

Power and Area-Efficient XNOR-AND Hybrid Binary Neural Networks Using TFT-Type Synaptic Device

IEEE ELECTRON DEVICE LETTERS

Lee, In-Seok; Kim, Hyeongsu; Park, Min-Kyu; Hwang, Joon; Koo, Ryun-Han; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 0

2023

Efficient Convolutional Processing of Spiking Neural Network With Weight-Sharing Filters

IEEE ELECTRON DEVICE LETTERS

Song, Seunghwan; Jeon, Bosung; Kim, Munhyeon; Kim, Jae-Joon

被引用次数: 2

2023

Channel-Length-Dependent Low-Frequency Noise Characteristics of Ferroelectric Junctionless Poly-Si Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Kim, Sangwoo; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Deok-Hwang; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2023

Optimizing Post-Metal Annealing Temperature Considering Different Resistive Switching Mechanisms in Ferroelectric Tunnel Junction

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Min, Kyung Kyu; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 10

2023

Unraveling Threshold Voltage Instability in Ferroelectric Junctionless FETs Using Low-Frequency Noise Measurement With Base Bias

IEEE ELECTRON DEVICE LETTERS

Shin, Wonjun; Koo, Ryun-Han; Kim, Sangwoo; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 2

2023

Comparative Analysis of n- and p-Type Ferroelectric Tunnel Junctions Through Understanding of Non-FE Resistance Switching

IEEE ELECTRON DEVICE LETTERS

Koo, Ryun-Han; Shin, Wonjun; Ryu, Sangwoo; Lee, Kyungmin; Park, Sung-Ho; Im, Jiseong; Ko, Jong-Hyun; Kim, Jeong-Hyun; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 6

2023

INSTA-BNN: Binary Neural Network with INSTance-aware Threshold

2023 IEEE/CVF INTERNATIONAL CONFERENCE ON COMPUTER VISION (ICCV 2023)

Lee, Changhun; Kim, Hyungjun; Park, Eunhyeok; Kim, Jae-Joon

被引用次数: 3

2023

Efficient Sampling and Grouping Acceleration for Point Cloud Deep Learning via Single Coordinate Comparison

2023 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Yoon, Hyunsung; Kim, Jae-Joon

被引用次数: 0

2023

1/f Noise in Synaptic Ferroelectric Tunnel Junction: Impact on Convolutional Neural Network

ADVANCED INTELLIGENT SYSTEMS

Shin, Wonjun; Min, Kyung Kyu; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 11

2023

V-LSTM: An Efficient LSTM Accelerator Using Fixed Nonzero-Ratio Viterbi-Based Pruning

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Kim, Taesu; Ahn, Daehyun; Lee, Dongsoo; Kim, Jae-Joon

被引用次数: 4

2023

In-Memory-Computed Low-Frequency Noise Spectroscopy for Selective Gas Detection Using a Reducible Metal Oxide

ADVANCED SCIENCE

Shin, Wonjun; Kim, Jaehyeon; Jung, Gyuweon; Ju, Suyeon; Park, Sung-Ho; Jeong, Yujeong; Hong, Seongbin; Koo, Ryun-Han; Yang, Yeongheon; Kim, Jae-Joon; Han, Seungwu; Lee, Jong-Ho

Self-Curable Synaptic Ferroelectric FET Arrays for Neuromorphic Convolutional Neural Network

ADVANCED SCIENCE

Shin, Wonjun; Im, Jiyong; Koo, Ryun-Han; Kim, Jaehyeon; Kwon, Ki-Ryun; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho; Kwon, Daewoong

被引用次数: 21

2023

Voltage scheme for string-select transistors to improve inhibition characteristics during 1-bit erase in vertical NAND flash

APPLIED PHYSICS LETTERS

Park, Sung-Ho; Yoo, Ho-Nam; Yang, Yeongheon; Back, Jong-Won; Koo, Ryun-Han; Kwon, Dongseok; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 2

2023

TAIM: Ternary Activation In-Memory Computing Hardware with 6T SRAM Array

PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022

Kang, Nameun; Kim, Hyungjun; Oh, Hyunmyung; Kim, Jae-Joon

被引用次数: 1

2022

Retention Improvement in Vertical NAND Flash Memory Using 1-bit Soft Erase Scheme and its Effects on Neural Networks

2022 INTERNATIONAL ELECTRON DEVICES MEETING, IEDM

Park, Sung-Ho; Kwon, Dongseok; Yoo, Ho-Nam; Back, Jong-Won; Hwang, Joon; Yang, Yeongheon; Kim, Jae-Joon; Lee, Jong-Ho

被引用次数: 11

2022

Variability analysis of ferroelectric FETs in program operation using low-frequency noise spectroscopy

APPLIED PHYSICS LETTERS

Shin, Wonjun; Bae, Jong-Ho; Kim, Jaehyeon; Koo, Ryun-Han; Kim, Jae-Joon; Kwon, Daewoong; Lee, Jong-Ho

被引用次数: 12

2022

Spiking Neural Network With Weight-Sharing Synaptic Array for Multi-input Processing

IEEE ELECTRON DEVICE LETTERS

Song, Seunghwan; Kim, Munhyeon; Jeon, Bosung; Ryu, Donghyun; Kim, Sihyun; Lee, Kitae; Lee, Jong-Ho; Kim, Jae-Joon; Shim, Wonbo; Kwon, Daewoong; Park, Byung-Gook

被引用次数: 3

2022

BitBlade: Energy-Efficient Variable Bit-Precision Hardware Accelerator for Quantized Neural Networks

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Ryu, Sungju; Kim, Hyungjun; Yi, Wooseok; Kim, Eunhwan; Kim, Yulhwa; Kim, Taesu; Kim, Jae-Joon

被引用次数: 25

2022

Extreme Partial-Sum Quantization for Analog Computing-In-Memory Neural Network Accelerators

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Kim, Yulhwa; Kim, Hyungjun; Kim, Jae-Joon

被引用次数: 3

2022

Real-Time Correlation Detection via Online Learning of a Spiking Neural Network with a Conductive-Bridge Neuron

ADVANCED ELECTRONIC MATERIALS

Kim, Dong-Won; Woo, Dae-Seong; Kim, Hea-Jee; Jin, Soo-Min; Jung, Sung-Mok; Kim, Dong-Eon; Kim, Jae-Joon; Shim, Tae-Hun; Park, Jea-Gun

被引用次数: 6

2022

Maximizing Parallel Activation of Word-Lines in MRAM-Based Binary Neural Network Accelerators

IEEE ACCESS

Ahn, Daehyun; Oh, Hyunmyung; Kim, Hyungjun; Kim, Yulhwa; Kim, Jae-Joon

被引用次数: 5

2021

Energy-efficient charge sharing-based 8T2C SRAM in-memory accelerator for binary neural networks in 28nm CMOS

IEEE ASIAN SOLID-STATE CIRCUITS CONFERENCE (A-SSCC 2021)

Oh, Hyunmyung; Kim, Hyungjun; Ahn, Daehyun; Park, Jihoon; Kim, Yulhwa; Lee, Inhwan; Kim, Jae-Joon

被引用次数: 3

2021

Mobileware: A High-Performance MobileNet Accelerator with Channel Stationary Dataflow

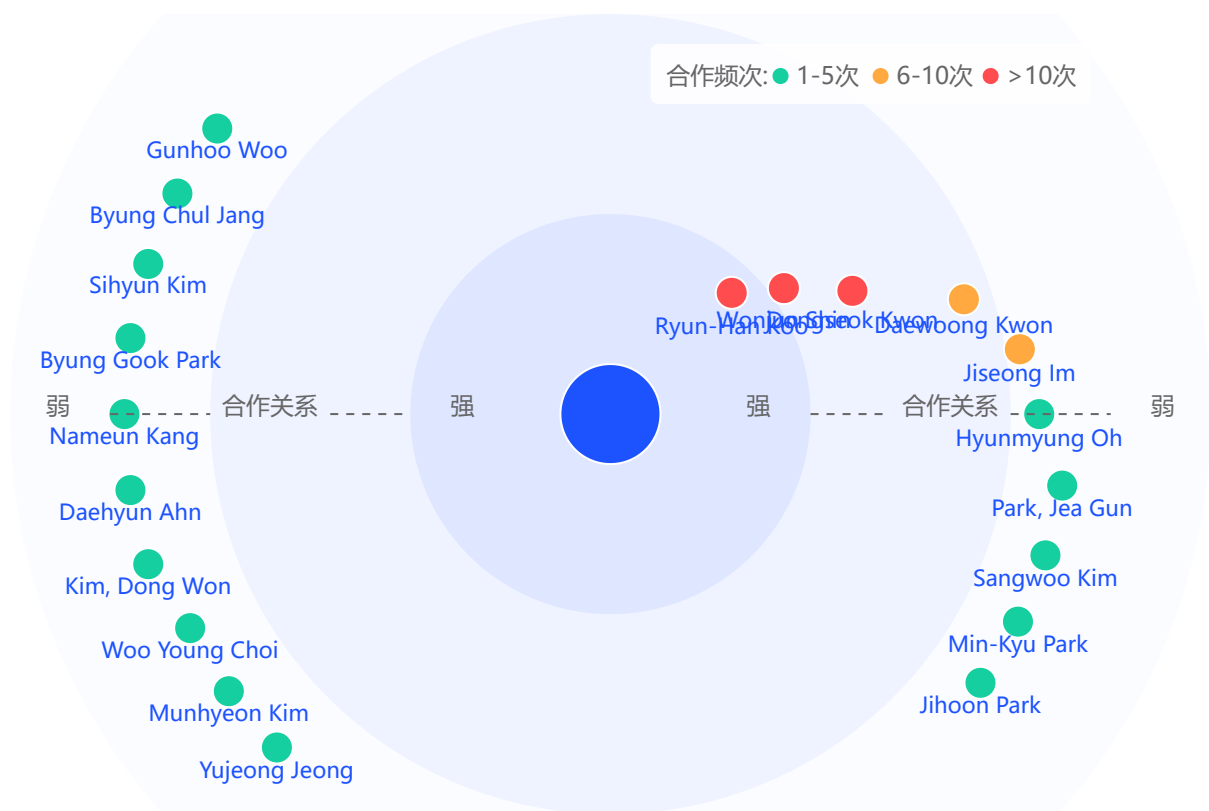
2021 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN (ICCAD)

Ryu, Sungju; Oh, Youngtaek; Kim, Jae-Joon

被引用次数: 6

2021

学术合作



Ryun-Han Koo

首尔国立大学

合作次数: 23

Wonjun Shin

成均馆大学

合作次数: 20

Dongseok Kwon

首尔国立大学

合作次数: 16

Daewoong Kwon

汉阳大学

合作次数: 9

Jiseong Im

首尔国立大学

合作次数: 6

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